MATHEMATICS (Senior Secondary) – PRACTICE TEST 2

Module IV: Co-ordinate Geometry (Lessons 13 – 16)

Cartesian System of Rectangular Co-ordinates · Straight Lines · Circles · Conic Sections

Time: 2 hours	**Maximum Marks:** 50
Name: _____	**Date:** _____

General Instructions: *(i) All questions are compulsory. (ii) Marks are shown against each question. (iii) Show all steps of your working. (iv) Use of a calculator is not permitted.*

SECTION A – Multiple Choice Questions (10 × 1 = 10 marks)

Choose the correct option.

1. The point which divides the join of (–3, 4) and (5, –2) externally in the ratio 3 : 1 is

(a) (1, 1) (b) (–5, 9) (c) (9, –5) (d) (–9, 5)

2. The acute angle between two lines with slopes 2 and 1/3 is

(a) 30° (b) 45° (c) 60° (d) 90°

3. The distance between the parallel lines $5x - 12y + 7 = 0$ and $5x - 12y - 19 = 0$ is

(a) 2 units (b) 1 unit (c) 13 units (d) 26 units

4. The radius of the circle $2x^2 + 2y^2 - 8x + 12y - 10 = 0$ is

(a) $\sqrt{2}$ (b) 3 (c) $3\sqrt{2}$ (d) 18

5. The equation of the circle of radius 4 which touches both the axes and lies in the first quadrant is

(a) $x^2 + y^2 - 8x - 8y - 16 = 0$ (b) $x^2 + y^2 - 8x - 8y + 16 = 0$ (c) $x^2 + y^2 + 8x + 8y + 16 = 0$ (d) $x^2 + y^2 - 4x - 4y + 16 = 0$

6. The directrix of the parabola $y^2 = -8x$ is

(a) $x = -2$ (b) $x = 2$ (c) $y = 2$ (d) $y = -2$

7. The eccentricity of the rectangular hyperbola $x^2 - y^2 = 16$ is

(a) $\sqrt{2}$ (b) 2 (c) $1/\sqrt{2}$ (d) $\sqrt{3}$

8. The foci of the ellipse $x^2/169 + y^2/144 = 1$ are

(a) $(\pm 5, 0)$ (b) $(0, \pm 5)$ (c) $(\pm 13, 0)$ (d) $(\pm 12, 0)$

9. The equation of the line which makes intercepts –3 and 2 on the x-axis and y-axis respectively is

(a) $2x + 3y - 6 = 0$ (b) $3x - 2y + 6 = 0$ (c) $2x - 3y + 6 = 0$ (d) $2x - 3y - 6 = 0$

10. When the origin is shifted to the point (2, 1) by translation of axes, the transformed equation of the line $2x - y + 3 = 0$ (in terms of the new co-ordinates x, y) is

(a) $2x - y + 3 = 0$ (b) $2x - y + 6 = 0$ (c) $2x - y - 6 = 0$ (d) $2x + y + 6 = 0$

SECTION B – Short Answer Questions I (6 × 2 = 12 marks)

11. Find the value of k for which the points (k, 3), (2, –1) and (–1, –5) are collinear.

12. Find the equation of the line which passes through the point (–1, 3) and makes equal intercepts on the co-ordinate axes.

13. Reduce the equation $\sqrt{3}\,x + y - 8 = 0$ to the normal (perpendicular) form and hence find the length of the perpendicular from the origin and the angle it makes with the positive direction of the x-axis.

14. Find the equation of the circle whose centre is (3, –1) and which passes through the point (6, 3).

15. Find the equation of the parabola whose vertex is at the origin, whose axis is the x-axis and which passes through the point (–2, 4).

16. Find the equation of the hyperbola (referred to its centre) whose foci are (±6, 0) and whose transverse axis is 8 units long.

SECTION C – Short Answer Questions II (4 × 4 = 16 marks)

17. Find the area of the quadrilateral ABCD whose vertices are A(1, 1), B(5, 2), C(6, 6) and D(2, 4).

18. Find the equations of the straight lines which pass through the point (1, 2) and make an angle of 45° with the line $2x - y + 3 = 0$.

19. Find the equation of the circle which passes through the points (5, 7) and (–2, 0) and whose centre lies on the line $2x - y = 1$. Also write down its radius.

20. The centre of an ellipse is at the origin and its major axis is along the x-axis. If its eccentricity is 3/5 and the length of its latus rectum is 32/5, find the equation of the ellipse and the co-ordinates of its foci.

SECTION D – Long Answer Questions (2 × 6 = 12 marks)

21. A parabola has its focus at S(2, 1) and the line $x + 2y + 1 = 0$ as its directrix.

(a) Find the equation of the parabola. [4]

(b) Find the equation of its axis. [1]

(c) Find the length of its latus rectum. [1]

22. The vertices of a triangle ABC are A(0, 6), B(–2, 2) and C(4, 0). Find

(a) the equation of the median through A, [2]

(b) the equation of the altitude through A, [2]

(c) the equation of the right bisector of the side BC, [1]

(d) the area of the triangle ABC. [1]

— End of Question Paper —

ANSWER KEY AND MARKING SCHEME

SECTION A

Q.	1	2	3	4	5	6	7	8	9	10
Ans.	(c)	(b)	(a)	(c)	(b)	(b)	(a)	(a)	(c)	(b)

Hints: Q1 (3·5 – 1·(–3))/2 = 9, (3·(–2) – 4)/2 = –5 ⇒ (9, –5). Q2 tan θ = (2 – 1/3)/(1 + 2/3) = 1. Q3 |7 + 19|/13 = 2. Q4 divide by 2: g = –2, f = 3, c = –5 ⇒ r = √18 = 3√2. Q5 centre (4, 4), r = 4 ⇒ (x–4)² + (y–4)² = 16. Q6 4a = 8, directrix x = a = 2. Q7 a = b ⇒ e = √2. Q8 a² = 169, b² = 144 ⇒ ae = √(169–144) = 5. Q9 x/(–3) + y/2 = 1 ⇒ 2x – 3y + 6 = 0. Q10 x = x′ + 2, y = y′ + 1 ⇒ 2x′ – y′ + 6 = 0.

SECTION B

11. Area = ½ |k(–1 + 5) + 2(–5 – 3) + (–1)(3 + 1)| = ½ |4k – 16 – 4| [1]. For collinearity area = 0 ⇒ 4k = 20 ⇒ **k = 5** [1]

12. Let the line be x/a + y/a = 1 ⇒ x + y = a [1]. Through (–1, 3): a = –1 + 3 = 2 ⇒ **x + y = 2** (i.e. x + y – 2 = 0) [1]

13. √(A²+B²) = √(3 + 1) = 2. Dividing by 2: (√3/2)x + (1/2)y = 4, i.e. x cos 30° + y sin 30° = 4 [1]. So **p = 4 units** and **α = 30°** [1]

14. r = distance between (3, –1) and (6, 3) = √(9 + 16) = 5 [1]. Equation: (x – 3)² + (y + 1)² = 25 ⇒ $\mathbf{x^2 + y^2 - 6x + 2y - 15 = 0}$ [1]

15. Axis along x-axis, vertex at origin ⇒ $y^2 = 4ax$ [½]. Point (–2, 4): 16 = 4a(–2) ⇒ 4a = –8 [1]. $\mathbf{y^2 = -8x}$ [½]

16. Standard form $x^2/a^2 - y^2/b^2 = 1$. Transverse axis 2a = 8 ⇒ a = 4; foci (±ae, 0) ⇒ ae = 6 [½]. b² = a²(e² – 1) = a²e² – a² = 36 – 16 = 20 [1]. $\mathbf{x^2/16 - y^2/20 = 1}$ [½]

SECTION C

17. Join AC to split ABCD into triangles ABC and ACD [1].

Area ABC = ½ |1(2 – 6) + 5(6 – 1) + 6(1 – 2)| = ½ |–4 + 25 – 6| = 15/2 [1½]

Area ACD = ½ |1(6 – 4) + 6(4 – 1) + 2(1 – 6)| = ½ |2 + 18 – 10| = 5 [1]

Area ABCD = 15/2 + 5 = **25/2 sq. units** [½]

18. Slope of 2x – y + 3 = 0 is $m_1 = 2$. Let the required slope be m. tan 45° = 1 = |(m – 2)/(1 + 2m)| [1].

Case 1: (m – 2)/(1 + 2m) = 1 ⇒ m – 2 = 1 + 2m ⇒ m = –3 [1]. Case 2: (m – 2)/(1 + 2m) = –1 ⇒ m – 2 = –1 – 2m ⇒ m = 1/3 [1].

Lines through (1, 2): y – 2 = –3(x – 1) ⇒ **3x + y – 5 = 0**; y – 2 = (1/3)(x – 1) ⇒ **x – 3y + 5 = 0** [1]

19. Let the circle be $(x - h)^2 + (y - k)^2 = r^2$, centre (h, k) with 2h – k = 1, so k = 2h – 1 [½].

Equal radii: (h – 5)² + (k – 7)² = (h + 2)² + k² [½]. Substituting k = 2h – 1: 5h² – 42h + 89 = 5h² + 5 ⇒ h = 2, k = 3 [2].

r² = (2 – 5)² + (3 – 7)² = 25. Equation: (x – 2)² + (y – 3)² = 25 ⇒ $\mathbf{x^2 + y^2 - 4x - 6y - 12 = 0}$; radius **5 units** [1]

20. $b^2 = a^2(1 - e^2) = a^2(1 - 9/25) = 16a^2/25$ [1]. Latus rectum $2b^2/a = 32a/25 = 32/5 \Rightarrow a = 5$, $b^2 = 16$ [1½].

Equation: $\mathbf{x^2/25 + y^2/16 = 1}$ [½]. Foci (±ae, 0) = **(±3, 0)** [1]

SECTION D

21. (a) For any point P(x, y) on the parabola, SP = distance from the directrix ⇒ $SP^2 = PM^2$ [1]:

$(x - 2)^2 + (y - 1)^2 = (x + 2y + 1)^2/5$ [1] ⇒ $5(x^2 - 4x + y^2 - 2y + 5) = x^2 + 4y^2 + 4xy + 2x + 4y + 1$ [1] ⇒ **$4x^2 - 4xy + y^2 - 22x - 14y + 24 = 0$** [1]

(b) Axis is through S and perpendicular to the directrix. Slope of directrix = –1/2, so slope of axis = 2: $y - 1 = 2(x - 2)$ ⇒ **$2x - y - 3 = 0$** [1]

(c) Distance of S from directrix = $|2 + 2 + 1|/\sqrt{5} = \sqrt{5}$. Latus rectum = 2 × (distance from focus to directrix) = **$2\sqrt{5}$ units** [1]

22. (a) Mid-point of BC = (1, 1). Slope of median = $(1 - 6)/(1 - 0) = -5$ ⇒ $y - 6 = -5x$ ⇒ **$5x + y - 6 = 0$** [2]

(b) Slope of BC = $(0 - 2)/(4 + 2) = -1/3$ ⇒ slope of altitude = 3. Through A(0, 6): $y = 3x + 6$ ⇒ **$3x - y + 6 = 0$** [2]

(c) The right bisector passes through the mid-point (1, 1) of BC with slope 3: $y - 1 = 3(x - 1)$ ⇒ **$3x - y - 2 = 0$** [1]

(d) Area = $\frac{1}{2}|0(2 - 0) + (-2)(0 - 6) + 4(6 - 2)| = \frac{1}{2}|12 + 16|$ = **14 sq. units** [1]